

Think of Triso fuel as a spherical capsule in an FHR (Ray, 090626)

Triso fuel in 3D: $\frac{\partial C}{\partial t} = D \nabla^2 C$ (Fick's 2nd Law)

$$\frac{\partial C}{\partial t} = D \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial C}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial C}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 C}{\partial \phi^2} \right]$$

Assume: (1) Triso fuel is uniform throughout (reduces dependency on θ, ϕ)

(2) Concentration of Triso fuel at any position (reduces the convection term) is equals to the concentration at the surface.

Triso fuel modelled as 1D equation: $\frac{\partial C}{\partial t} = D \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial C}{\partial r} \right) \right]$

Using product rule to simplify:

$$\begin{aligned} \frac{1}{r^2} \left[\frac{\partial}{\partial r} \left(r^2 \frac{\partial C}{\partial r} \right) \right] &= \frac{1}{r^2} \left[2r \frac{\partial C}{\partial r} + r^2 \frac{\partial^2 C}{\partial r^2} \right] \\ &= \frac{2}{r} \frac{\partial C}{\partial r} + \frac{\partial^2 C}{\partial r^2} \end{aligned}$$

Simplified Triso Equation: $\frac{\partial C}{\partial t} = D \left(\frac{\partial^2 C}{\partial r^2} + \frac{2}{r} \frac{\partial C}{\partial r} \right)$

Discretisation method: FTCS (Forward Time Centred Space)

Using Central Difference Method to discretise RHS:

$$\frac{\partial^2 C}{\partial r^2} = \frac{C_{i-1}^j - 2C_i^j + C_{i+1}^j}{(\Delta r)^2}$$

$$\frac{\partial C}{\partial r} = \frac{C_{i+1}^j - C_{i-1}^j}{2\Delta r}$$

Using Forward Euler Method to discretise LHS:

$$\frac{\partial C}{\partial t} = \frac{C_i^{j+1} - C_i^j}{\Delta t}$$

Final Discretised Equation:

$$\frac{C_i^{j+1} - C_i^j}{\Delta t} = D \left(\frac{C_{i+1}^j - 2C_i^j + C_{i-1}^j}{(\Delta r)^2} + \frac{2}{r} \frac{C_{i+1}^j - C_{i-1}^j}{2\Delta r} \right)$$

Making C_i^{j+1} the subject of the Equation:

$$\begin{aligned} C_i^{j+1} &= C_i^j + \frac{D\Delta t}{(\Delta r)^2} (C_{i+1}^j - 2C_i^j + C_{i-1}^j) + \frac{D\Delta t}{r\Delta r} (C_{i+1}^j - C_{i-1}^j) \\ &= C_i^j \left(1 - 2 \frac{D\Delta t}{(\Delta r)^2} \right) + C_{i+1}^j \left(\frac{D\Delta t}{(\Delta r)^2} + \frac{D\Delta t}{(\Delta r)^2} \cdot \frac{1}{i} \right) + C_{i-1}^j \left(\frac{D\Delta t}{(\Delta r)^2} - \frac{D\Delta t}{(\Delta r)^2} \cdot \frac{1}{i} \right) \end{aligned}$$

Initial conditions:

$$\frac{\partial C}{\partial t} = 0$$

Boundary conditions: Assume $i=1$ to $i=5$ (4 shells, 5 nodes)

For Centre node $i=1$:

$$\left. \frac{\partial C}{\partial r} \right|_0 = 0 \text{ (symmetry)}$$

$$\frac{\partial C}{\partial r} = \frac{C_2 - C_0}{2\Delta r} \text{ (Central Difference Method)}$$

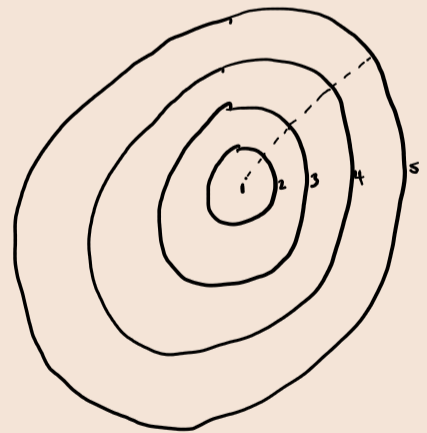
$$0 = \frac{C_2 - C_0}{2\Delta r}$$

$$C_0 = C_2$$

$$\frac{\partial^2 C}{\partial r^2} = \frac{C_2 - 2C_1 + C_0}{(\Delta r)^2}$$

$$= \frac{C_2 - 2C_1 + C_2}{(\Delta r)^2}$$

$$= \frac{2C_2 - 2C_1}{(\Delta r)^2}$$



RHS simplification:

$$\frac{\partial^2 C}{\partial r^2} + \frac{2}{r} \frac{\partial C}{\partial r} = \frac{\partial^2 C}{\partial r^2} + \lim_{r \rightarrow 0} \frac{2}{r} \frac{\partial C}{\partial r} \quad (\text{L'Hopital rule})$$

$$= \frac{\partial^2 C}{\partial r^2} + 2 \frac{\partial^2 C}{\partial r^2}$$

Q1.

$$= 3 \frac{\partial^2 C}{\partial r^2}$$

← Should this be final Equation to be discretised or should I use previous time from

$$\frac{\partial C}{\partial t} = D \left(3 \frac{\partial^2 C}{\partial r^2} \right)$$

$$\frac{C_{i+1}^j - C_i^j}{\Delta t} = D(3) \left(\frac{2C_1 - 2C_0}{(\Delta r)^2} \right)$$

$$C_i^{j+1} = C_i^j + 6D\Delta t \left(\frac{C_1 - C_0}{(\Delta r)^2} \right) \quad [\text{Concentration @ Node 0 at next timestep}]$$

For Outer node $i=5$:

$$\left. \frac{\partial C}{\partial r} \right|_0 = 0 \quad (\text{insulation})$$

$$\frac{\partial C}{\partial r} = \frac{C_6 - C_4}{2\Delta r}$$

$$0 = \frac{C_6 - C_4}{2\Delta r}$$

$$C_6 = C_4$$

$$\frac{\partial^2 C}{\partial r^2} = \frac{C_6 - 2C_5 + C_4}{(\Delta r)^2}$$

$$= \frac{2C_4 - 2C_5}{(\Delta r)^2}$$

$$\frac{\partial C}{\partial t} = D \left(3 \frac{\partial^2 C}{\partial r^2} \right)$$

$$\frac{C_{i+1}^j - C_i^j}{\Delta t} = D(3) \left(\frac{2C_4 - 2C_5}{(\Delta r)^2} \right)$$

$$C_i^{j+1} = C_i^j + 6D\Delta t \left(\frac{C_4 - C_5}{(\Delta r)^2} \right) \quad [\text{Concentration @ Node 4 at next timestep}]$$

For concentration @ Node 2, 3, 4 at next timestep:

Node 2: $\frac{1}{i} = \frac{1}{2}$ Use for loop in code.

Node 3: $\frac{1}{i} = \frac{1}{3}$

Node 4: $\frac{1}{i} = \frac{1}{4}$

Node 2:

$$C_i^{j+1} = C_i^j \left(1 - 2 \frac{D\Delta t}{(\Delta r)^2} \right) + C_{i+1}^j \left(\frac{D\Delta t}{(\Delta r)^2} + \frac{D\Delta t}{(\Delta r)^2} \cdot \frac{1}{2} \right) + C_{i-1}^j \left(\frac{D\Delta t}{(\Delta r)^2} - \frac{D\Delta t}{(\Delta r)^2} \cdot \frac{1}{2} \right)$$

Node 3:

$$C_i^{j+1} = C_i^j \left(1 - 2 \frac{D\Delta t}{(\Delta r)^2} \right) + C_{i+1}^j \left(\frac{D\Delta t}{(\Delta r)^2} + \frac{D\Delta t}{(\Delta r)^2} \cdot \frac{1}{3} \right) + C_{i-1}^j \left(\frac{D\Delta t}{(\Delta r)^2} - \frac{D\Delta t}{(\Delta r)^2} \cdot \frac{1}{3} \right)$$

Node 4:

$$C_i^{j+1} = C_i^j \left(1 - 2 \frac{D\Delta t}{(\Delta r)^2} \right) + C_{i+1}^j \left(\frac{D\Delta t}{(\Delta r)^2} + \frac{D\Delta t}{(\Delta r)^2} \cdot \frac{1}{4} \right) + C_{i-1}^j \left(\frac{D\Delta t}{(\Delta r)^2} - \frac{D\Delta t}{(\Delta r)^2} \cdot \frac{1}{4} \right)$$

Plot the concentration from $i=1$ to $i=5$ at each timestep.

Layer each graph together.

Final Triiso Equation for simple diffusion :

$$\frac{\partial C}{\partial t} = D \left[\frac{C_{i-1} - 2C_i + C_{i+1}}{(\Delta r)^2} + \frac{2}{r} \frac{C_{i+1} - C_{i-1}}{2\Delta r} \right]$$

$$= D \left[\frac{r(C_{i-1} - 2C_i + C_{i+1}) + \Delta r(C_{i+1} - C_{i-1})}{r(\Delta r)^2} \right]$$

$$\boxed{\frac{\partial C}{\partial t} = D \left[\frac{C_{i-1}(r - \Delta r) + C_i(-2r) + C_{i+1}(r + \Delta r)}{r(\Delta r)^2} \right]}$$

Convert into code to plot

Base case 1 : Diffusivity D is constant

Base case 2 : Diffusivity D varies with Temperature T .

Possible source terms to add :

① Fission product : $\Sigma_f \phi \gamma$ where Σ_f = macroscopic fission cross section
 ϕ = neutron flux at r
 γ = fission product yield

② Radioactive Decay : $-\lambda C$ where $\lambda = \frac{\ln 2}{t_{1/2}}$

C = concentration

③ Grain Boundary Trapping : $-K C$ where K = net rate of fission products being trapped & unable to diffuse

C = concentration

Additional considerations :

① D is also different for each layer of Triiso fuel for given components

② D changes with irradiation damage over time

$$\frac{\partial C}{\partial t} = 0 \left[\frac{C_{i-1}(r-\Delta r) + C_i(-2r) + C_{i+1}(r+\Delta r)}{r(\Delta r)^2} \right]$$

To make C_{i+1} the subject:

$$\frac{dC}{dt} \cdot \frac{r(\Delta r)^2}{D} = C_{i-1}(r-\Delta r) + C_i(-2r) + C_{i+1}(r+\Delta r)$$

$$\frac{\frac{dC}{dt} \cdot \frac{r(\Delta r)^2}{D} - C_{i-1}(r-\Delta r) - C_i(-2r)}{r+\Delta r} = C_{i+1}$$

$$\frac{dC}{dt} = \frac{C_{i+1} - C_{i-1}}{2\Delta t} \quad \times \quad \frac{dC}{dt} = \frac{C_{i+1} - C_i}{\Delta t}$$

$$\frac{C_{i+1} - C_{i-1}}{2\Delta t} \cdot \frac{r(\Delta r)^2}{D} - C_{i-1}(r-\Delta r) - C_i(-2r) = C_{i+1}$$

$$\frac{C_{i+1} - C_{i-1}}{2\Delta t} = D \frac{C_{i-1}(r-\Delta r)}{r(\Delta r)^2}$$

$$+ D \frac{C_i(-2r)}{r(\Delta r)^2}$$

$$+ D \frac{C_{i+1}(r+\Delta r)}{r(\Delta r)^2}$$

LHS:

$$\frac{C_{i+1}}{2\Delta t} - \frac{C_{i-1}}{2\Delta t}$$

Subject:

$$\frac{C_{i+1}}{2\Delta t} - D \frac{C_{i+1}(r+\Delta r)}{r(\Delta r)^2} = \frac{C_{i-1}}{2\Delta t} + D \frac{C_{i-1}(r-\Delta r)}{r(\Delta r)^2} + D \frac{C_i(-2r)}{r(\Delta r)^2}$$

$$C_{i+1} \left(\frac{1}{2\Delta t} - \frac{D(r+\Delta r)}{r(\Delta r)^2} \right) - C_{i-1} \left(\frac{1}{2\Delta t} + \frac{D(r-\Delta r)}{r(\Delta r)^2} \right) + C_i \left(\frac{D(-2r)}{r(\Delta r)^2} \right)$$

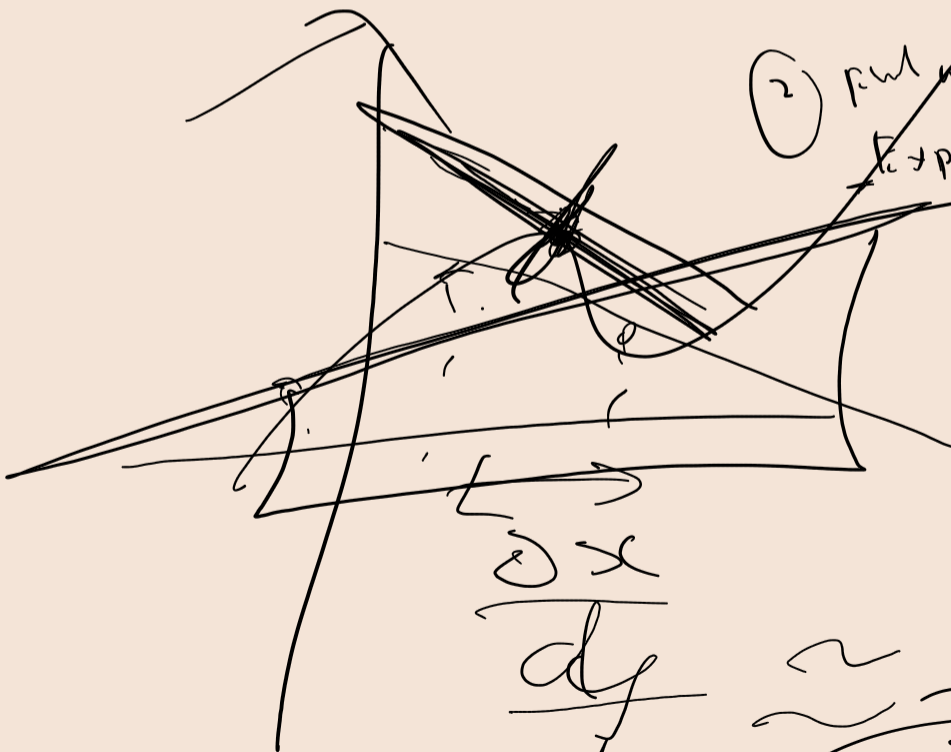
$$C_{i+1} = \frac{1}{\frac{1}{2\Delta t} - \frac{D(r+\Delta r)}{r(\Delta r)^2}} \left[\frac{C_{i-1}}{2\Delta t} + \frac{D(r-\Delta r)}{r(\Delta r)^2} C_{i-1} + C_i \left(\frac{D(-2r)}{r(\Delta r)^2} \right) \right]$$

$$r = i \Delta r$$



i = number of shells

r = total radius



(1) Find out how FEM
can go wrong in graph +
like

(2) Find out how to curve
Exponent Error

(3) Redo case

(4) Watch Youtube
vid on FEM

$$\frac{\partial x}{\partial y} \approx \frac{\Delta y}{\Delta x}$$

$$\approx \frac{\Delta y}{\Delta x}$$

Course
Nikolaev

Approach through:

- ① Focus on Base case + ignore source terms
- ② Find cases to benchmark solutions against

$$x^{n+1} = x^n + h f(x^{n+1}, t^{n+1})$$